INTRODUCTION TO SMITH'S METHODS

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Lecture 1

1. Overview of Arithmetic Statistics

We begin with the following, very influential, conjecture in arithmetic statistics.

Conjecture 1.1 (Cohen–Lenstra, 1984)**.** *Let p be an odd prime and let A be a finite abelian p-group. Then*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ \mathrm{Cl}_K[p^\infty] \cong A\}}{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X\}} = \frac{\prod_{i=1}^{\infty}(1-p^{-i})}{\#\mathrm{Aut}(A)}.$$

There is very little proved about this conjecture, but a lot of computational evidence has been found and it is strongly believed. Indeed, it is not known for a single pair (p, A)! The main thing that is known is the following average result.

Theorem 1.2 (Davenport–Heilbronn, 1970s)**.**

$$\sum_{\substack{K/\mathbb{Q} \\ |D_K|\le X}} \#\mathrm{Cl}_K[3] \sim cX$$

and the leading constant matches the average predicted by the Cohen–Lenstra distribution.

Conjecture 1.3 (Cohen–Lenstra–Gerth)**.** *Let A be a finite abelian 2-group. Then*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ 2\mathrm{Cl}_K[2^\infty] \cong A\}}{\#\{K/\mathbb{Q} \ : \text{imag. quad. } \ |D_K| \le X\}} = \frac{\prod_{i=1}^{\infty}(1-2^{-i})}{\#\mathrm{Aut}(A)}$$

The idea here is that in this setting, $\mathrm{Cl}_K[2]$ is special. It is "predictable" and in fact quite large ($\#\mathrm{Cl}_K[2] = 2^{\omega(D_K)-1}$ where ω is the number of prime divisors). Multiplying by 2 kills this contribution, and the claim is somehow that this is the only predictable behaviour present.

The first result on this is the work of Fouvry–Klülners on the 4-torsion.

Theorem 1.4 (Fouvry–Klüners, 2007)**.** *We have for all $n \ge 0$ that*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ 2\mathrm{Cl}_K[4] \cong \mathbb{F}_2^n\}}{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X\}} = \lim_{r\to\infty} P_{\mathrm{Mat}}(r,n)$$

where $P_{\mathrm{Mat}}(r,n)$ is the probability that a uniformly selected matrix $M \in \mathrm{Mat}(r \times r, \mathbb{F}_2)$ has kernel of dimension n.

Then Alex Smith was able to get from 4 to 8 torsion under GRH.

Theorem 1.5 (Smith, 2016)**.** *Assume GRH. Then for all $m \geq j \geq 0$ we have*

$$\lim_{X\to\infty} \frac{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \ 2\mathrm{Cl}_K[8] \cong \mathbb{Z}/2\mathbb{Z}^{m-j} \oplus (\mathbb{Z}/4\mathbb{Z})^j\right\}}{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \ 2\mathrm{Cl}_K[4] \cong (\mathbb{Z}/2\mathbb{Z})^m\right\}} = P_{\mathrm{Mat}}(m,j)$$

In 2017, Smith generalised this: there is a filtration of $\mathbb{F}_2$-vector spaces

$$\mathrm{Cl}_K[2] \supseteq 2\mathrm{Cl}_K[4] \supseteq 4\mathrm{Cl}_K[8] \supseteq 8\mathrm{Cl}_K[16] \supseteq \dots,$$

and he studies the behaviour of this filtration. It is a worthwhile exercise to do, to convince yourself that this filtration recovers the isomorphism class of the abelian group. Note also that $2^k\mathrm{Cl}_K[2^{k+1}] = \mathrm{Cl}_K[2] \cap 2^k\mathrm{Cl}_K$.

Definition 1.6. We write $r_{i,K} := \dim_{\mathbb{F}_2} 2^{i-1}\mathrm{Cl}_K[2^i]$.

Smith proves the following.

Theorem 1.7 (Smith, 2017)**.** *Let $k \geq 2$ and let $r_{k+1} \leq r_k \leq \dots \leq r_2$. Then*

$$\lim_{X\to\infty} \frac{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \quad r_{i,K} = r_i \quad \forall 2 \leq i \leq k+1\right\}}{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \quad r_{i,K} = r_i \quad \forall 2 \leq i \leq k\right\}} = P_{\mathrm{Mat}}(r_{k+1}, r_k)$$

The idea here is that you have some Markov process, so that you don't care about the history of the process beyond the position you're on!

Remark 1.8. This implies Conjecture 1.3 (see K.–Pagano, JEMS, §14). Note that in the setting of Theorem 1.5 we have $r_2 = m$ and $r_3 = j$.

Since this work of Smith:

- Generalisation to number fields (under roots of unity assumptions) and quadratic twists of abelian varieties (Smith 2022).
- Counting squarefree $1 \leq d \leq X$ such that $x^2 - dy^2 = -1$ has a solution $x, y \in \mathbb{Z}$ (K.–Pagano 2022)
- Upper and lower bounds (conditional on a converse theorem) on how often as n varies there are solutions of $x^3 + y^3 = n$ for $x, y \in \mathbb{Q}$ (K.–Smith 2024).
- BSD implies Goldfeld's conjecture (Smith 2025).

2. Overview of the lecture series

Our goal in this series is to do the following.

- Sketch a proof of theorem 1.
- Give an unconditional proof of Theorem 1.5 using a new tool from recent work of K.–Smith.
- Time permitting, we might also see how to access Theorem 1.7 in general, but that might be difficult to reach.

The idea is that our proof of Theorem 1.5 will be easier than the original, though it makes use of the many developments since then and is still not terribly easy.

Now for an underlying theme that we will see repeatedly. Accessing the class group directly is very hard, but what we end up being able to do is to *compare* class groups. A promising start for this is as follows. Denote by $H_K(2)$ the maximal abelian unramified extension of K with $\mathrm{Gal}(H_K(2)/K)$ being an $\mathbb{F}_2$-vector space. Then

(1) we can give an explicit description of $H_K(2)$ (next lecture!)

(2) for two different imaginary quadratic fields K, K' there is an interesting intersection $H_K(2) \cap H_{K'}(2)$ (i.e. bigger than $\mathbb{Q}$) which completely fails if 2 is replaced by an odd prime.

The following gives a good mental example to keep in mind when we're talking about comparing class groups, but we will need to do something more general.

Theorem 2.1 (Stevenhagen)**.** *Let $p \equiv 1 \mod 8$ and let χ_p be the quadratic dirichlet character associated to p and let ζ_8 be an element of order 8 in $(\mathbb{Z}/p\mathbb{Z})^\times$. Write $h_d := \#\mathrm{Cl}_{\mathbb{Q}(\sqrt{d})}$ and when $d > 0$ write $h_d^+ := \#\mathrm{Cl}^+_{\mathbb{Q}(\sqrt{d})}$ for the narrow class number. Then*

$$\begin{aligned} 8 \mid h_{-p} &\iff \chi_p(1+\zeta_8^2) = 1 \\ 8 \mid h_{-2p} &\iff \chi_p(1+\zeta_8^2)\chi_p(\zeta_8) = 1 \\ 8 \mid h_{2p}^+ &\iff \chi_p(1+\zeta_8^2) = \chi(\zeta_8) = 1 \end{aligned}$$

Now suppose that $\chi(1+\zeta_p^2) = \chi(\zeta_8) = 1$. If $\chi(1+\zeta_8) = 1$,

$$16 \mid h_{2p}^+ \iff 16 \mid h_{-p} \text{ and } 16 \mid h_{-2p}$$

Alternatively if $\chi(1+\zeta_8) = -1$ then

$$16 \mid h_{2p}^+ \iff 16 \mid h_{-2p} \text{ and } 8 \mid\mid h_{-p}$$

The slogan here is that for 16-ranks, the relationships between h_{2p}^+, h_{-2p}, h_{-p} is governed by a splitting condition.

Lecture 2

3. Genus theory and cocycles

Notation 3.1. Let $N := \mathbb{Q}_2/\mathbb{Z}_2 = \varinjlim \mathbb{Z}/2^k\mathbb{Z}$. We view N as a (continuous) $G_{\mathbb{Q}}$-module via the discrete topology and trivial action. For each $\chi \in \mathrm{Hom}(G_{\mathbb{Q}}, \{\pm 1\})$ we define $N(\chi)$ to be N as an abelian group but with the $G_{\mathbb{Q}}$-action for all $\sigma \in G_{\mathbb{Q}}$ and $n \in N$

$$\sigma(n) = \chi(\sigma)n.$$

Recall that squarefree numbers are in bijection with quadratic extensions of $\mathbb{Q}$ and hence with elements of $\mathrm{Hom}(G_{\mathbb{Q}}, \{\pm 1\})$. If x is a squarefree number then we define the character χ_x to be the image of x in $\mathrm{Hom}(G_{\mathbb{Q}}, \{\pm 1\})$, which is precisely the map $\sigma \mapsto \frac{\sigma(\sqrt{x})}{\sqrt{x}}$. We write $N(x) := N(\chi_x)$.

Theorem 3.2. *There is an exact sequence for each $k \geq 1$ and $K = \mathbb{Q}(\sqrt{x})$*

$$0 \to \mathbb{Z}/2^k\mathbb{Z} \to Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k]) \to \mathrm{Cl}_K^{\vee}[2^k] \to 0,$$

where Z^1 denotes the group of 1-cocycles, H_K is the Hilbert class field, and $\vee$ denotes dual.

Proof. We assume the following fact from exercise sheet 2: For every quadratic field $K/\mathbb{Q}$, the Hilbert class field H_K is Galois over $\mathbb{Q}$ and

$$1 \to \mathrm{Gal}(H_K/K) \to \mathrm{Gal}(H_K/\mathbb{Q}) \to \mathrm{Gal}(K/\mathbb{Q}) \to 1$$

is split with $\mathrm{Gal}(H_K/\mathbb{Q}) \cong \mathrm{Gal}(H_K/K) \rtimes \mathrm{Gal}(K/\mathbb{Q})$. Writing $\mathrm{Gal}(K/\mathbb{Q}) = \langle \sigma \rangle$, then the generator σ acts by inversion on $\mathrm{Gal}(H_K/K)$. Moreover by the Artin map $\mathrm{Gal}(H_K/K) \cong \mathrm{Cl}_K$ and this is the natural action of $\mathrm{Gal}(K/\mathbb{Q})$ on Cl_K.

This theorem will essentially follow from inflation restriction, but we need to be quite careful as we must work on the level of cocycles. There is for each $k \geq 1$ and each $K = \mathbb{Q}(\sqrt{x})$ a natural restriction map

$$\begin{aligned} Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k]) \xrightarrow{\mathrm{res}} & Z^1(\mathrm{Gal}(H_K/K), N(x)[2^k]) \\ &= \mathrm{Hom}(\mathrm{Gal}(H_K/K), N(x)[2^k]) \\ &= \mathrm{Hom}(\mathrm{Cl}_K, \mathbb{Z}/2^k\mathbb{Z}) \\ &\cong \mathrm{Cl}_K^{\vee}[2^k] \end{aligned}$$

Lemma 3.3. res *is surjective.*

Proof. Take $\phi \in \mathrm{Hom}(\mathrm{Gal}(H_K/K), N(x)[2^k])$. By the fact we are assuming above, $\mathrm{Gal}(H_K/\mathbb{Q}) \cong \mathrm{Gal}(H_K/K) \rtimes \mathrm{Gal}(K/\mathbb{Q})$. We define $\psi \in \mathrm{Map}(\mathrm{Gal}(H_K/\mathbb{Q}),\ N(x)[2^k])$ by $\psi = (\phi, 0)$ when written in this semidirect product decomposition. Indeed, ψ is a cocycle, since

$$\begin{aligned} \psi((a_1, g_1)(a_2, g_2)) &= \psi(a_1 + \chi_x(g_1)a_2,\ g_1g_2) \\ &= \phi(a_1) + \chi_x(g_1)\phi(a_2) \\ &= g_1 \cdot_{\chi_x} \psi(a_2, g_2) + \psi(a_1, g_1) \end{aligned}$$

$\square$

Using this, it is now sufficient to see that the kernel of res is a copy of $\mathbb{Z}/2^k\mathbb{Z}$. The kernel is

$$\ker(\mathrm{res}) = \inf(Z^1(\mathrm{Gal}(K/\mathbb{Q}), N(x)[2^k])) \cong \mathbb{Z}/2^k\mathbb{Z},$$

via evaluation at $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. This we leave as an exercise!

[R:This can be seen as follows: let $\phi \in \ker(\mathrm{res})$. Then representing $\mathrm{Gal}(H_K/\mathbb{Q})$ as the semidirect product in the usual way we have $\phi(\sigma,\tau) = \phi((\sigma,0)\cdot(0,\tau)) = (\sigma,0)\cdot\phi(0,\tau) = \phi(0,\tau)$. Hence $\phi \in \inf(Z^1(\mathrm{Gal}(K/\mathbb{Q}), N(x)[2^k])$. The converse is immediate, so the first equality holds. Note that inf is injective on 1-cocycles by definition.

Elements of $Z^1(\mathrm{Gal}(K/\mathbb{Q}), N(x)[2^k])$ are functions $f : \mathrm{Gal}(K/\mathbb{Q}) \to N(x)[2^k]$ which satisfy the cocycle relations, which here (since $\mathrm{Gal}(K/\mathbb{Q})$ is so small) are just given by $0 = f(0) = f(\sigma^2) = \sigma\cdot f(\sigma) + f(\sigma) = (1-\chi_x(\sigma))f(\sigma)$. But since $1-\chi_x(\sigma) = 0$ this just means we can choose any image for σ. Hence there are 2^k elements and they are identified by their image under evaluation at σ.] $\square$

Definition 3.4. Let $Z^1_{\mathrm{nr}}(G_{\mathbb{Q}}, N(x)[2^k]) = \inf\left(Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k])\right)$ for $K = \mathbb{Q}(\sqrt{x})$. Note that this unramified decoration really depends on x in general!

These are precisely the cocycles which restrict to $\mathrm{Hom}_{\mathrm{nr}}(G_K, N(x)[2^k]) \cong \mathrm{Cl}_K^\vee[2^k]$, i.e. those that restrict trivially to $\mathrm{Hom}(I_v, N(x)[2^k])$ for all places v where I_v is a choice of inertia subgroup at v.

Theorem 3.5 (Gauss genus theory)**.** *Let $K = \mathbb{Q}(\sqrt{-x})$ with $x \equiv -1 \mod 4$ a positive squarefree integer. Then for each odd prime p let $p^* = \pm p$ with choice of sign so that $p^* \equiv 1 \mod 4$. Then as homomorphism groups we have an equality*

$$Z^1_{\mathrm{nr}}(G_{\mathbb{Q}}, N(-x)[2]) = \langle \chi_{p^*} \ : \ p \mid x \rangle,$$

where the χ_{p^*} are all linearly independet over $\mathbb{Q}$ for ramification reasons. In particular

$$\#\mathrm{Cl}_K[2] = 2^{\omega(D_K)-1}$$

via Theorem 3.2.

Proof. We cheat slightly. $\mathrm{Hom}(G_{\mathbb{Q}}, \{\pm 1\})$ has a basis given by the χ_{p^*} for p odd primes, χ_{-1} and χ_2. We will check for each basis element whether it is unramified, which is not sufficient, but the general argument isn't much harder.

Indeed consider the field diagram

$$\begin{tikzcd}
 & K(\sqrt{p^*}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{-x/p^*}) \arrow[dr, dash] & K = \mathbb{Q}(\sqrt{-x}) \arrow[d, dash] & \mathbb{Q}(\sqrt{p^*}) \arrow[dl, dash] \\
 & \mathbb{Q} &
\end{tikzcd}$$

If $p \mid x$ then we want to show that χ_{p^*} is unramified on restriction to G_K. Then note that $\mathbb{Q}(\sqrt{p^*})/\mathbb{Q}$ is ramified only at p, and so $K(\sqrt{p^*})/K$ is ramified at most at primes above p. But what about at p? Since p is odd, we get that $\mathbb{Q}(\sqrt{-x/p^*})$ is unramified over p. Hence the total ramification degree is 2 and $K(\sqrt{p^*})/K$ is unramified.

In the second case, if $p \nmid x$ then we want $K(\sqrt{p^*})/K$ to be ramified at p, which is similar. $\square$

Lecture 3

4. The Artin Pairing and the Rédei matrix

Recall that we were planning to understand the behaviour of our class group Cl_K by studying the steps in the filtration

$$\mathrm{Cl}_K[2] \supseteq 2\mathrm{Cl}_K[4] \supseteq 4\mathrm{Cl}_K[8] \supseteq \dots$$

Since Genus theory will describe the group at the top, our job today will be to study the transition between the steps in this filration. This will make use of the Artin pairing.

Definition 4.1 (Artin pairing)**.** For a finite abelian 2-group A we write $A^\vee = \mathrm{Hom}(A, \mathbb{C}^\times)$. We then define pairings

$$\mathrm{Art}_k : 2^{k-1}A[2^k] \times 2^{k-1}A^\vee[2^k] \to \{\pm 1\}$$

given simply by mapping a pair (a, χ) to $\psi(a)$ where $\psi \in A^\vee[2^k]$ is such that $2^{k-1}\psi = \chi$.

Of course, we need to know that this is well defined.

Lemma 4.2. Art_k *is well defined, i.e. it does not depend on the choice of lift* ψ.

Proof. Let us take a pair (a, χ) in the domain. Let ψ_1, ψ_2 be two lifts of χ to $A^\vee[2^k]$, so that $2^{k-1}(\psi_1 - \psi_2) = 0$. But if is $b \in A[2^k]$ is a lift of a then

$$\psi_1(a) = \psi_1(a) - \psi_2(a) + \psi_2(a) = (\psi_1 - \psi_2)(2^{k-1}b) + \psi_2(a) = \psi_2(a).$$

□

Why is this pairing useful to us? Well it will tell us precisely which elements lift to the next stage of the filtration.

Lemma 4.3. *The left kernel of* Art_k *is* $2^kA[2^{k-1}]$ *and the right kernel is* $2^kA^\vee[2^{k+1}]$.

Proof. By the independence of lifting in the definition of Art_k, really Art_k arises from the pairing $2^{k-1}A[2^k] \times A^\vee[2^k]$ given by evaluation. In particular, the left kernel pairs trivially with all of $A^\vee[2^k]$ under the evaluation (or canonical) pairing

$$\begin{aligned} A \times A^\vee &\to \mathbb{C}^\times \\ (a, \chi) &\mapsto \chi(a). \end{aligned}$$

It is elementary to show that the orthogonal complement of $A^\vee[2^k]$ under this canonical pairing is 2^kA. Hence the left kernel of Art is $(2^{k-1}A[2^k]) \cap 2^kA = 2^kA[2^{k+1}]$. □

Now we must relate this finite abelian group theory back to class groups, and explain why we decided to call this the Artin pairing.

Lemma 4.4. *Identifying* $\mathrm{Cl}_K \cong \mathrm{Gal}(H_K/K)$ *via class field theory as before, in the commutative diagram*

$$\begin{tikzcd} \mathrm{Cl}_K[2] \times \mathrm{Cl}_K^\vee[2] \arrow[r, "\mathrm{Art}_1"] & \{\pm 1\} \\ \mathrm{Cl}_K[2] \times \mathrm{Hom}(\mathrm{Gal}(H_K/K), \{\pm 1\}) \arrow[u, "\sim"] \arrow[r, "\mathrm{Art}_{K,1}"] & \{\pm 1\} \end{tikzcd}$$

the bottom row map is given on the class of an unramified prime ideal $\mathfrak{p}$ *and character* χ *by*

$$(\mathfrak{p}, \chi) \mapsto \chi\left(\mathrm{Frob}_{\mathfrak{p}}\right)$$

where $\mathrm{Frob}_{\mathfrak{p}}$ *is Frobenius.*

Now, recall from genus theory that if $-x \equiv 1 \mod 4$ and $x = p_1 \dots p_r$ for distinct odd primes p_i then the map

$$\varphi_x : \mathbb{F}_2^r \to \mathrm{Cl}_K[2]$$

given by $(e_1, \dots, e_r) \mapsto [\prod_{i=1}^r \mathfrak{p}_i^{e_i}]$ where $\mathfrak{p}_i$ are the primes above each p_i. We saw this is surjective with 1-dimensional kernel $(1, 1, \dots, 1)$.

We similarly have a map

$$\psi_x : \mathbb{F}_2^r \to \mathrm{Cl}_K^\vee[2]$$

given by $(e_1, \dots, e_r) \mapsto \prod_{i=1}^r \chi_{p^*}^{e_i}$. Again this is surjective with kernel $(1, \dots, 1)$ as before. We're then going to define a lift of $\mathrm{Art}_{K,1}$ as follows.

Definition 4.5. We define $\widetilde{\mathrm{Art}_{K,1}} : \mathbb{F}_2^r \times \mathbb{F}_2^r \to \{\pm 1\}$ to be the lift of the Artin pairing under φ_x and ψ_x. That is,

$$\widetilde{\mathrm{Art}_{K,1}}(a, b) = \mathrm{Art}_{K,1}(\varphi_x(a), \psi_x(b)).$$

Definition 4.6. Let $-x \equiv 1 \mod 4$ be squarefree, $K = \mathbb{Q}(\sqrt{-x})$, and $r = \omega(x)$. Then the Rédei matrix R_x of K is defined to be the $r \times r$ matrix with entries $\widetilde{\mathrm{Art}_{K,1}}(\mathbf{e}_i, \mathbf{e}_j)$ where the $\mathbf{e}_i$ are the elementary basis vectors with 1 in the ith component and zeros everywhere else. [R:Note that this is precisely the matrix such that in the standard basis,

$$\widetilde{\mathrm{Art}_{K,1}}(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T R_x \mathbf{w}$$

]

Theorem 4.7. *Let* $-x \equiv 1 \mod 4$ *be squarefree,* $K = \mathbb{Q}(\sqrt{-x})$*, and* $r = \omega(x)$*. Then*

$$\dim_{\mathbb{F}_2} 2\mathrm{Cl}_K[4] = r - 1 - \mathrm{rk}(R_x).$$

Proof. Use that the left kernel of $\mathrm{Art}_{K,1}$ is $2\mathrm{Cl}_K[4]$ and that $\mathbb{F}_2^r \xrightarrow{\varphi_x} \mathrm{Cl}_K[2]$ is surjective with 1 dimensional kernel. □

We now want to determine the entries in the Rédei matrix in terms of symbols we are already comfortable with.

Theorem 4.8. *Write* $R_x(i, j)$ *for the* i, j *entry in* R_x*. Then we have for all indices* $1 \leq i, j \leq r$

$$R_x(i, j) = \begin{cases} \left(\frac{p_j^*}{p_i}\right) & \text{if } i \neq j \\ \sum_{k \neq i} R_x(i, k) & \text{if } i = j \end{cases}$$

where we interpret Legendre symbols as taking values in $\mathbb{F}_2$*.*

Proof. In the case $i \neq j$, by Lemma 4.4 and the definition of $\widetilde{\mathrm{Art}_{K,1}}$, we have

$$\begin{aligned} \widetilde{\mathrm{Art}_{K,1}}(\mathbf{e}_i, \mathbf{e}_j) &= \chi_{p_j^*}|_{G_K}(\mathrm{Frob}(\mathfrak{p}_i)) \\ &= \left(\frac{p_j^*}{p_i}\right) \end{aligned}$$

Note that this second equality is decided by whether the residue degree of p_i is 1 or 2 in the (unramified!) extension $K(\sqrt{p_j^*})/K$ where $K = \mathbb{Q}(\sqrt{-x})$ as before, hence the second equality.

For the case $i = j$, note that $\prod_{j=1}^{r} \chi_{p_j^*} = \chi_{-x}$ is trivial when restricted to G_K, so all rows have sum equal to 1. $\square$

Hence we can write down R_x as the following matrix (taking Legendre symbols to be valued in $\mathbb{F}_2$)

$$R_x = \begin{pmatrix} * & \left(\frac{p_2^*}{p_1}\right) & \cdots & \left(\frac{p_r^*}{p_1}\right) \\ \left(\frac{p_1^*}{p_2}\right) & * & \cdots & \left(\frac{p_r^*}{p_2}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{p_1^*}{p_r}\right) & \left(\frac{p_2^*}{p_r}\right) & \cdots & * \end{pmatrix} \tag{4.1}$$

where the $*$ are filled in by the sum of the rest of their row.

Example 4.9. Let us consider $x = 3 \times 5 \times 13 = 195$. Then

- $\mathrm{Cl}_K^\vee[2] = \langle \chi_{-3}, \chi_5, \chi_{13} \rangle$
- $\mathrm{Cl}_K[2] = \langle \mathfrak{p}_3, \mathfrak{p}_5, \mathfrak{p}_{13} \rangle$

and hence the Rédei matrix is

$$R = \begin{pmatrix} * & \left(\frac{5}{3}\right) & \left(\frac{13}{3}\right) \\ \left(\frac{-3}{5}\right) & * & \left(\frac{13}{5}\right) \\ \left(\frac{-3}{13}\right) & \left(\frac{5}{13}\right) & * \end{pmatrix} = \begin{pmatrix} * & 1 & 0 \\ 1 & * & 1 \\ 0 & 1 & * \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

This has rank 2 and so we get $\dim \mathrm{Cl}_K[4] = 3 - 1 - 2 = 0$.

Lecture 4

A good heuristic then for $\mathrm{Art}_{K,1}$ will be that it should be a random matrix with entries in $\mathbb{F}_2$ (or ± 1), and that $\widetilde{\mathrm{Art}_{K,1}}$ should be a random matrix in $\mathbb{F}_2$ subject to the constraint that $(1,\dots,1)$ is in the left and right kernel (i.e. the sums of the rows and columns are zero).

5. Gridding

[R:Our goal is now to let x range among positive squarefree integers and understand $\dim_{\mathbb{F}_2} 2\mathrm{Cl}_{\mathbb{Q}(\sqrt{-x})}[4]$. For convenience, we have restricted to only the odd x with $-x \equiv 1 \mod 4$. By Theorem 4.7, we want to understand the average behaviour of $\omega(x) - \mathrm{rk}(R_x)$ as x varies. A natural way to want to do this is to look at integers with exactly r prime factors for each r individually (freezing the value of $\omega(x)$, and the dimension of R_x). Then we would like to let the prime factors of x vary at random, and use analytic techniques to describe the variation of the Legendre symbols in the entries of R_x to show that these matrices equidistribute among appropriate matrices in $\mathrm{Mat}(r \times r, \mathbb{F}_2)$. Grids are how we make this precise.]

We now allow ourselves a parameter H, which will be large and eventually tend to infinity.

Definition 5.1. Define $\alpha_j := e^{(\log\log H)^{100}}(1+\frac{1}{\log H})^j$. For $H \geq 100$, a *grid of height H* is a pair

$$(X, i_{\mathrm{small}})$$

where $X = X_1 \times \dots X_r$ is a product such that

(1) Each X_i is a subset of the primes.
(2) For $i \leq i_{\mathrm{small}}$, we ensure $|X_i| = 1$ and moreover the element $p_i \in X_i$ should satisfy $p_i < e^{(\log\log H)^{100}}$.
(3) For $i > i_{\mathrm{small}}$ we insist that X_i is all of the primes in the interval $[\alpha_j, \alpha_{j+1})$ for some j.
(4) For all $1 \leq i < j \leq r$, then for every $p_i \in X_i$ and $p_j \in X_j$ we have $p_i < p_j$.
(5) Every $x = (p_1,\dots,p_r) \in X$ satisfies $p_1 p_2 \dots p_r \leq H$

We view a grid X as a subset of squarefree integers according to their prime factorisations. That is, a tuple $(p_1,\dots,p_r) \in X$ corresponds to the squarefree integer $p_1 \dots p_r$.

Theorem 5.2. *We have $X \cap X' = \emptyset$ for two distinct grids X, X' of height H.*

Proof. Exercise! □

Definition 5.3. We call a grid $X = X_1 \times \dots \times X_r$ good if

(1) $\frac{4}{5}\log\log H \leq r \leq (\log\log H)^2$
(2) $i_{\mathrm{small}} \leq (\log\log\log H)^2$
(3) $\#\left\{i \ : \ X_i \subseteq \left[1, e^{\sqrt{\log H}}\right)\right\} \leq \frac{2}{3}\log\log H$

What is this saying? Why is this a good notion of good? Well using Erdős–Kac type heuristics: we expect that for a random squarefree integer n we should have about $\log\log y$ prime divisors smaller than y. (1) is concerned with the total number of prime factors, which by the heuristic should be around $\log\log H$, so these bounds describe a 'typical' behaviour. Now (2) is concerned with the number of small prime divisors, those smaller than $e^{(\log\log H)^{100}}$. The number of such primes

should be around $\log\log\left(e^{(\log\log H)^{100}}\right) = 100\log\log\log H$. Certainly our bound should then model reality. Finally (3) is bounds the number of prime divisors which are of medium size, meaning smaller than $e^{\sqrt{\log H}}$. This should be around $\log\log(e^{\sqrt{\log H}}) = \frac{1}{2}\log\log H$, so we bound by $\frac{2}{3}\log\log H$.

Now we show that one really can restrict to studying integers in good grids.

Theorem 5.4. $\#\left\{n \le H \ : \ \text{sqf}\,, n \notin \bigcup_{X \text{ good}} X\right\} \ll \frac{H}{\log\log\log H}$

Proof. We break this proof into cases. Firstly we will deal with the number of integers which are not in any grid at all, then we deal with those who are not in a *good* grid.

Case 1 (not in a grid): Firstly, we bound those $n \le H$ which are not in any grid of height H at all. This happens only if one of the following is true.

(1) n has a large neighbour who excludes the grid. That is, for example, say there is an X_i such that the $p_i \mid n$ from that X_i has a neighbour $q \in X_i$ such that $m = qn/p_i > H$. In this case, we would have thrown the grid out. In this case, checking the definitions, we would have

$$n \ge H(1 + \frac{1}{\log H})^{-\omega(n)}.$$

As $\omega(n) \ll \frac{\log H}{\log\log H}$ the last condition implies that $n \ge He^{-c/\log\log H}$ for some $c > 0$. So there are at lost $H/\log\log H$ such integers, which is within our proposed bound.

(2) n has two distinct prime divisors in an interval $[\alpha_j, \alpha_{j+1})$. The number of such n is bounded by

$$\begin{aligned}\sum_{j \ll (\log H)^2} \sum_{p,q \in [\alpha_j, \alpha_{j+1})} \frac{H}{pq} &\ll H \sum_{j \ll (\log H)^2} \sum_{p,q \in [\alpha_j, \alpha_{j+1})} \frac{1}{pq} \\ &\ll H \sum_{j \ll (\log H)^2} \left(\sum_{p \in [\alpha_j, \alpha_{j+1})} \frac{1}{p} \right)^2\end{aligned}$$

Applying Mertens theorem, $\sum_{p \le x} p^{-1} = \log\log X + M + O(e^{-\sqrt{\log X}})$ for some constant M. So in particular, our bound becomes

$$\begin{aligned}&H \sum_{j \ll (\log H)^2} (\log\log\alpha_{j+1} - \log\log\alpha_j)^2 \\ &= H \sum_{j \ll (\log H)^2} \log\left(\frac{\log\alpha_{j+1}}{\log\alpha_j}\right)^2 \\ &= H \sum_{j \ll (\log H)^2} \log\left(\frac{\log\alpha_j + \log(1 + \frac{1}{\log H})}{\log\alpha_j}\right)^2 \\ &\ll H \sum_{j \ll (\log H)^2} \left(\frac{1}{\log\alpha_j \log H}\right)^2\end{aligned}$$

which is smaller than our bound once again.

Case 2 (in a grid, but not a good one): We now need to deal with integers which are in a grid, but not a good one.

Definition 5.5. Now we call a squarefree integer n good if

(1) $\frac{4}{5}\log\log H \leq \omega(n) \leq (\log\log H)^2$
(2) $\#\{p \mid n \ : \ p < \alpha_0\} \leq (\log\log\log H)^2$
(3) $\#\left\{p \mid n \ : \ p \leq 2e^{\sqrt{\log H}}\right\} \leq \frac{2}{3}\log\log H$

In particular, if n is good and in a grid X then it must be in a good grid. It suffices to estimate n failing the conditions (1), (2), or (3). We begin with the first.

Lemma 5.6. *For $X > 3$, we have*

$$\sum_{n\leq X}\omega(n) = X\log\log X + O(X)$$
$$\sum_{n\leq X}\omega(n)^2 = X(\log\log X)^2 + O(X\log\log X)$$

In particular, $\sum_{n\leq X}(\omega(n) - \log\log X)^2 \ll X\log\log X$.

Proof. We apply Mertens theorem.

$$\sum_{n\leq X}\omega(n) = \sum_{p\leq X}\sum_{\substack{n\leq X\\ p\mid n}} 1 = \sum_{p\leq X}\left(\frac{X}{p} + O(1)\right) = O(\frac{X}{\log X}) + X\sum_{p\leq X}\frac{1}{p} = X\log\log X + O(X)$$

Then we compute

$$\begin{aligned}\sum_{n\leq X}\omega(n)^2 &= \sum_{p,q\leq X}\sum_{\substack{n\leq X\\ p\mid n\\ q\mid n}} 1 = O(X\log\log X) + \sum_{p,q\leq X}\sum_{\substack{n\leq X\\ pq\mid n}} 1 = O(X\log\log X) + \sum_{p,q\leq X}\left\lfloor\frac{X}{pq}\right\rfloor\\ &= O(X\log\log X) + \sum_{p,q\leq X}\frac{X}{pq}\\ &= O(X\log\log X) + X\left(\sum_{p\leq X}\frac{1}{p}\right)^2\\ &= X(\log\log X)^2 + O(X\log\log X).\end{aligned}$$

For the final claim, we expand and note that the leading terms cancel and so we are bounded by the error. □

Lemma 5.7. *The number of squarefree $n < H$ which fail to be good because of (1) in Definition 5.5 is $\ll H/\log\log H$*

Proof. We apply Lemma 5.6, in particular we know a bound on the second moment

$$\sum_{n\leq H}(\omega(n) - \log\log H)^2 \ll H\log\log H. \tag{5.1}$$

Note that the contribution of any element of $\left\{n \leq X \ : \ \omega(n) \leq \frac{4}{5}\log\log H\right\}$ to the sum (5.1) is $\gg (\log\log H)^2$ and so

$$\#\left\{n \leq H \ : \ \omega(n) \leq \frac{4}{5}\log\log H\right\} \ll \frac{1}{(\log\log H)^2}\sum_{n\leq H}(\omega(n) - \log\log H)^2 \ll H/\log\log H.$$

Similarly, the contribution of any element of $\{n \le X \ : \ \omega(n) > (\log\log H)^2\}$ to the second moment sum is $\gg (\log\log H)^4$ and so similarly we obtain

$$\#\{n \le X \ : \ \omega(n) > (\log\log H)^2\} \ll H/(\log\log H)^3.$$

Then we get that the number of n failing condition (1) is small as required □

The argument for bounding the number who fail (2) or (3) is similar, and left as an exercise. □

Lecture 5

6. The large sieve

The most well-known version of the large sieve is due to Heath-Brown, which we now state below.

Theorem 6.1 (Heath-Brown)**.** *Let $M, N \in \mathbb{Z}_{\ge 1}$, and let $\beta_1, \dots, \beta_N \in \mathbb{C}$ with $|\beta_j| \le 1$. Then for all $\varepsilon > 0$*

$$\sum_{m \le M} \left| \sum_{\substack{n \le N \\ \text{odd}}} \beta_n \left(\frac{m}{n}\right) \right| \ll_\varepsilon \frac{(MN)^{1+\varepsilon}}{\min\{M^{1/2}, N^{1/2}\}}$$

What really makes this result possible is the double sum, that you have two axes of randomness. Note that on the left there are MN terms, so we're seeing almost square-root saving in one of the variables.

Example 6.2. Take $M = N = X$ and $\beta_n = 1_{\text{prime}}$. Then

$$\sum_{n \le X} \left| \sum_{\substack{p \le X \\ \text{prime}}} \left(\frac{m}{n}\right) \right| \ll_\varepsilon X^{3/2+\varepsilon},$$

where the Legendre symbol is really telling us about the splitting of p in $\mathbb{Q}(\sqrt{m})$. From Markov's inequality, for m outside of an exceptional set $\mathcal{E}$ with $|\mathcal{E}| \le X^{1-\varepsilon/2}$ (i.e. for almost all m)

$$\left| \sum_{\substack{p \le X \\ \text{prime}}} \left(\frac{m}{n}\right) \right| \ll X^{\frac{1+\varepsilon}{2}}$$

is the GRH bound for $\mathbb{Q}(\sqrt{m})$.

The proof of Heath-Brown's result is quite technical, and we won't have time to get into it in this series, so we'll look at this slightly weaker but more pliable result of Friedlander–Iwaniec.

Theorem 6.3 (Friedlander–Iwaniec)**.** *Let $M, N \ge 1$ and $\beta_1, \dots, \beta_N \in \mathbb{C}$ with $|\beta_i| \le 1$. Then for all $\varepsilon > 0$*

$$\sum_{\substack{m \le M \\ \text{odd}}} \left| \sum_{\substack{n \le N \\ \text{odd}}} \beta_n \left(\frac{m}{n}\right) \right| \ll_\varepsilon \frac{(MN)^{1+\varepsilon}}{\min\{M^{1/6}, N^{1/6}\}}$$

The disadvantage of this result is that the saving is not as good, but the advantage is that it is easier and generalises well to settings like cubic characters, Hecke characters over number fields (see work of Lemke Oliver–Smith), and similar.

Before the theorem, let us prove a lemma.

Lemma 6.4.

$$\sum_{\substack{m\leq M\\ \text{odd}}}\left|\sum_{\substack{n\leq N\\ \text{odd}}}\beta_n\left(\frac{m}{n}\right)\right| \ll_\varepsilon MN^{1/2+\varepsilon}+M^{1/2}N^2$$

Remark 6.5. This is only nontrivial (meaning beating the trivial bound of MN) when $MN\geq M^{1/2}N^2$. Checking the algebra, this is when $M\geq N^2$.

Remark 6.6. From now on in lectures we will make substantial use of the Cauchy–Schwarz inequality:

$$\left|\sum_{a\leq A}x_ay_a\right|\leq\left(\sum_{x\leq A}x_a^2\right)^{1/2}\left(\sum_{y\leq A}y_a^2\right)^{1/2}.$$

Try not to forget it!

Proof. Applying Cauchy–Schwarz with $x_a=1$ and $y_a=\left|\sum_{\substack{n\leq N\\ \text{odd}}}\beta_n\left(\frac{m}{n}\right)\right|$, we have

$$\begin{aligned}\sum_{\substack{m\leq M\\ \text{odd}}}1\cdot\left|\sum_{\substack{n\leq N\\ \text{odd}}}\beta_n\left(\frac{m}{n}\right)\right| &\leq\left(\sum_{\substack{m\leq M\\ \text{odd}}}1\right)^{1/2}\left(\sum_{\substack{m\leq M\\ \text{odd}}}\left|\sum_{\substack{n\leq N\\ \text{odd}}}\beta_n\left(\frac{m}{n}\right)\right|^2\right)^{1/2}\\ &\leq M^{1/2}\left(\sum_{m\leq M}\sum_{\substack{n_1,n_2\leq N\\ \text{odd}}}\beta_{n_1}\overline{\beta_{n_2}}\left(\frac{m}{n_1n_2}\right)\right)^{1/2}\end{aligned}$$

Now switching the order of summation and applying the triangle inequality we obtain

$$\begin{aligned}\sum_{\substack{n_1,n_2\leq N\\ \text{odd}}}\left|\sum_{m\leq M}\left(\frac{m}{n_1n_2}\right)\right| &= \sum_{\substack{n_1,n_2\leq N\\ \text{odd}\\ n_1n_2=\square}}\left|\sum_{m\leq M}\left(\frac{m}{n_1n_2}\right)\right|+\sum_{\substack{n_1,n_2\leq N\\ \text{odd}\\ n_1n_2\neq\square}}\left|\sum_{m\leq M}\left(\frac{m}{n_1n_2}\right)\right|\\ &\ll N(\log N)M+N^4\end{aligned}$$

where for the first term we have applied the trivial bound (since the character is trivial so there is no saving to be had!). For the second term we used that this is a non-principal Dirichlet character of modulus at most N^2 so when we sum over N^2 integers in a row we get zero. Collecting terms, we get the result. □

Proof of Theorem 6.3. We claim that this theorem is equivalent to the following statement: for all α_m,β_n in $\mathbb{C}$ with absolute value at most 1, we have

$$\left|\sum_{\substack{m\leq M\\ \text{odd}}}\sum_{n\leq N}\alpha_m\beta_n\left(\frac{m}{n}\right)\right|\ll_\varepsilon\frac{(MN)^{1+\varepsilon}}{\min\left\{M^{1/6},N^{1/6}\right\}}. \tag{6.1}$$

The equivalence is as follows.

FI $\implies$ (6.1) Just apply the triangle inequality in (6.1) to get

$$\left|\sum_{\substack{m\le M\\ \text{odd}}}\sum_{n\le N}\alpha_m\beta_n\left(\frac{m}{n}\right)\right| \le \sum_{\substack{m\le M\\ \text{odd}}}\left|\sum_{\substack{n\le N\\ \text{odd}}}\alpha_m\beta_n\left(\frac{m}{n}\right)\right|$$

Then pull the α_m out and conclude.

(6.1) $\implies$ **FI** Take

$$\alpha_m = \begin{cases} \frac{\left|\sum_{n\le N}\beta_n\left(\frac{m}{n}\right)\right|}{\sum_{n\le N}\beta_n\left(\frac{m}{n}\right)} & \text{if defined} \\ 0 & \text{else} \end{cases}$$

and apply (6.1).

Now we prove (6.1), for which a main tool will be Hölder's inequality.[R:add to remark later] WLOG we assume that $M \ge N$ and by the triangle inequality it suffices to bound

$$\sum_{\substack{n\le N\\ \text{odd}}} 1\cdot\left|\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right|$$

By Hölder this is at most

$$\left(\sum_{\substack{n\le N\\ \text{odd}}} 1\right)^{2/3}\left(\sum_{\substack{n\le N\\ \text{odd}}}\left|\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right|^3\right)^{1/3} \le N^{2/3}\left(\sum_{\substack{n\le N\\ \text{odd}}}\left|\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right|^3\right)^{1/3}$$

We let S now denote the remaining double sum here, and set

$$\gamma_n = \frac{\left|\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right|^3}{\left(\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right)^3},$$

so that $|\gamma_n|\le 1$ and

$$S = \sum_{\substack{n\le N\\ \text{odd}}}\gamma_n\left(\sum_{\substack{m\le M\\ \text{odd}}}\alpha_m\left(\frac{m}{n}\right)\right)^3.$$

We then expand the cube to obtain

$$\begin{aligned}
S &= \sum_{\substack{n\leq N\\ \text{odd}}} \gamma_n \sum_{\substack{m_1,m_2,m_3\leq M\\ \text{odd}}} \alpha_{m_1}\alpha_{m_2}\alpha_{m_3}\left(\frac{m_1m_2m_3}{n}\right)\\
&\leq \sum_{\substack{m_1,m_2,m_3\leq M\\ \text{odd}}} 1\cdot\left|\sum_{\substack{n\leq N\\ \text{odd}}}\gamma_n\left(\frac{m_1m_2m_3}{n}\right)\right|\\
&\leq \sum_{\substack{\ell\leq M^3\\ \text{odd}}}\tau_3(\ell)\cdot\left|\sum_{\substack{n\leq N\\ \text{odd}}}\gamma_n\left(\frac{\ell}{n}\right)\right|\\
&\ll M^{\varepsilon}\sum_{\substack{\ell\leq M^3\\ \text{odd}}}\left|\sum_{\substack{n\leq N\\ \text{odd}}}\gamma_n\left(\frac{\ell}{n}\right)\right|.
\end{aligned}$$

Now we apply Lemma 6.4 and conclude. □